Name: [redacted]

Roll # CT-24020

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR(Computer Science)
FALL SEMESTER EXAMINATIONS 2025

Time : 3 Hours | **Batch 2024** | **Dated : 09-DEC-25** | **Max Marks : 60**

Data Structure Algorithms & Applications - CT-159

Note: Attempt all question.

1. **Explain** the differences among: Full Binary Tree, Complete Binary Tree and Perfect Binary Tree. **Illustrate** with an example diagram for each. [CLO-02, Marks-05]
2. Write the **algorithm** for searching a value in **Binary Search Tree** and **Analyse** its **best, worst and average case complexities.** [CLO-02, Marks-05]
3. **Insert** the following keys in order into an **AVL Tree: 30, 20, 40, 10, 25, 50, 5. Apply** rotations wherever required and draw the final **AVL tree.** Show complete working for each insertion of node. [CLO-01, Marks-05]
4. **Write pseudo-code** for a function that evaluates a postfix expression but also detects and report common errors:

   a) Too many operands b) Too many operators c) Division by zero d) Invalid characters.

   The function should return the result if valid, or a specific error code otherwise. **Test** your algorithm with: "**5 2 + 3 *" (valid)** and "**5 2 + +" (invalid).** [CLO-01, Marks-05]
5. Given the **adjacency matrix** of weighted graph, draw the graph and apply **Dijkstra's Algorithm** to find the shortest path from source vertex "3" to all other vertices. [CLO-01, Marks-05]

V	0	1	2	3	4	5	6
0	0	3	0	0	0	0	0
1	3	0	8	0	0	0	0
2	0	8	0	7	0	5	0
3	0	0	7	0	9	15	4
4	0	0	0	9	0	10	0
5	0	0	5	15	10	0	7
6	0	0	0	4	0	7	0

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6. Given the **adjacency matrix data** in question-5, draw the graph and use **Prim's Algorithm** to find the Minimum Spanning Tree (MST) starting from vertex "**5**". [CLO-01, Marks-05]

7. You are given a directed acyclic weighted graph. **Discover a linearized sorted order** by applying the **topological sort algorithm**. Also, write the **pseudo-code** of the algorithm. [CLO-01, Marks-05]

Vertex	Neighbor : Edge Weight
A	E:4 → H:12 → NULL
B	C:5 → NULL
C	NULL
D	A:7 → H:3 → NULL
E	F:6 → NULL
F	B:2 → I:4 → C:6 → G:9 → NULL
G	NULL
H	E:8 → I:11 → NULL
I	J:3 → NULL
J	G:2 → NULL

8. Create a hash table step-by-step using double hashing and hash functions **h(k)=k mod 13** and **h'(k)= 3 + k mod 9** when inserting keys **17, 12, 4, 1, 36, 25, 29, 6**. [CLO-01, Marks-10]

9. **Evaluate** two different pivot selection strategies for **Quick Sort** (e.g., first element vs. median-of-three). **Discuss** how each strategy impacts the overall time complexity and justify which is generally considered better for average performance. [CLO-02, Marks-05]

10. **Argue** whether the following modification still constitutes a "**Linear Search**": "Check the first element, then the last, then the second, then the second-last, and so on, moving inwards." Justify your answer based on the fundamental defining characteristics of the algorithm (order of checking, worst-case scenario, and time complexity). [CLO-02, Marks-10]

———Best of Luck———

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